Zinan Lin

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Employment

Microsoft Research — **Redmond, WA, USA**
Senior Researcher — *Oct. 2022-Present*

Education

Carnegie Mellon University — **Pittsburgh, PA, USA**
Ph.D., Department of Electrical and Computer Engineering — *2017–2023*
Secondary Master in Machine Learning, School of Computer Science
Secondary Master in Electrical and Computer Engineering
Ph.D. advisors: Giulia Fanti and Vyas Sekar
Grade: 4.0/4.0

Tsinghua University — **Beijing, China**
Bachelor of Engineering, Department of Electronic Engineering — *2013–2017*
Grade: 92/100. Rank: 5/195

Honors and Awards

2nd Place (Out of 2211 Teams) in AI Mathematical Olympiad 2, with Yichen You, Xuefei Ning, https://www.kaggle.com/competitions/ai-mathematical-olympiad-progress-prize-2/leaderboard. Solution: https://github.com/imagination-research/aimo2/ — *2025*

12th Annual Best Scientific Cybersecurity Paper, with Boxin Wang, Weixin Chen, Hengzhi Pei, Chulin Xie, Mintong Kang, Chenhui Zhang, Chejian Xu, Zidi Xiong, Ritik Dutta, Rylan Schaeffer, Sang T. Truong, Simran Arora, Mantas Mazeika, Dan Hendrycks, Yu Cheng, Sanmi Koyejo, Dawn Song, Bo Li, granted by the National Security Agency (NSA), https://www.nsa.gov/Press-Room/News-Highlights/Article/Article/3990748/nsa-awards-authors-of-assessment-of-trustworthiness-in-gpt-models/ — *2024*

Best Machine Learning and Security Paper (Cybersecurity Best Paper Award), with Boxin Wang, Weixin Chen, Hengzhi Pei, Chulin Xie, Mintong Kang, Chenhui Zhang, Chejian Xu, Zidi Xiong, Ritik Dutta, Rylan Schaeffer, Sang T. Truong, Simran Arora, Mantas Mazeika, Dan Hendrycks, Yu Cheng, Sanmi Koyejo, Dawn Song, Bo Li, granted by SpringerOpen, https://cybersecurity.springeropen.com/award-2024 — *2024*

ICML Spotlight (Top 3.5%), with Chulin Xie, Arturs Backurs, Sivakanth Gopi, Da Yu, Huseyin Inan, Harsha Nori, Haotian Jiang, Huishuai Zhang, Yin Tat Lee, Bo Li, Sergey Yekhanin — *2024*

NeurIPS Outstanding Paper & Oral (Top 0.045%), with Boxin Wang, Weixin Chen, Hengzhi Pei, Chulin Xie, Mintong Kang, Chenhui Zhang, Chejian Xu, Zidi Xiong, Ritik Dutta, Rylan Schaeffer, Sang T. Truong, Simran Arora, Mantas Mazeika, Dan Hendrycks, Yu Cheng, Sanmi Koyejo, Dawn Song, Bo Li *2023*

NeurIPS Oral (Synthetic Data Generation with Generative AI Workshop), with Sivakanth Gopi, Janardhan Kulkarni, Harsha Nori, Sergey Yekhanin *2023*

Top Reviewers (Top 2.6%) in NeurIPS 2023, https://nips.cc/Conferences/2023/ProgramCommittee *2023*

AAAI Scholarship, granted by Association for the Advancement of Artificial Intelligence *2022*

Outstanding Reviewer (Top 8%) in NeurIPS 2021, https://nips.cc/Conferences/2021/ProgramCommittee *2021*

IMC Best Paper Finalist, with Alankar Jain, Chen Wang, Giulia Fanti, Vyas Sekar *2020*

Top Reviewers (Top 7.6%) in ICML 2020, https://icml.cc/Conferences/2020/Reviewers *2020*

Cylab Presidential Fellowship, granted by Carnegie Mellon University *2020*

Siemens FutureMakers Fellowship, granted by Siemens *2019*

Best Reviewers (Top 400) in NeurIPS 2019, https://nips.cc/Conferences/2019/Reviewers *2019*

NeurIPS Spotlight (Top 4.1%), with Kiran Thekumparampil, Ashish Khetan, and Sewoong Oh *2018*

Presidential Fellowship, granted by Carnegie Mellon University *2017*

Carnegie Institute of Technology Dean's Fellow, granted by Carnegie Mellon University *2017*

Outstanding Bachelor Thesis, granted by Tsinghua University *2017*

Meritorious Winner (9% Worldwide), COMAP's Math Contest in Modeling *2015, 2016, 2017*

National Scholarship, granted by the government of China *2014, 2015, 2016*

Tsinghua Spark Class Fellowship (Top 1%), for top students on scientific research *2015*

The First Prize, National Physics Contest for College Student *2014*

Experience

NVIDIA (Research Intern) **Santa Clara, CA, USA**
Host: Ming-Yu Liu, Xun Huang *May 2021–Dec. 2021*
- Topic: Denoising Diffusion Probabilistic Models (DDPM)

Google (Research Intern) **Mountain View, CA, USA**
Host: Yundi Qian *May 2020–Aug. 2020*
- Topic: compiler optimizations with reinforcement learning

Microsoft Research Asia (Research Intern) **Beijing, China**
Managers: Fei Gao, Taifeng Wang *Mar. 2017–Jun. 2017*
- Topic: a large-scale empirical study of optimization methods

Luogu Website (Cofounder and Developer) **China**
https://www.luogu.com.cn/ *2013*
- One of the biggest online judges in China.

Skills

Programming Languages

C, C++, Python, Java, (Visual) Basic, Pascal, Haskell, MATLAB, Mathematica, PHP, JavaScript, HTML, CSS, SQL, Verilog, Assembly, bash, shell, LaTeX, etc.

Machine Learning Frameworks

TensorFlow, PyTorch, Theano, Keras, Blocks, CNTK, etc.

Teaching Assistant

CMU 18752: Estimation, Detection and Learning — **Pittsburgh, PA, USA**
Instructor: Rohit Negi — *Spring 2020, Spring 2021*

Services

I am serving as a reviewer/PC member for:

Conference on Neural Information Processing Systems (NeurIPS)	*2019, 2020, 2021, 2022, 2023, 2025*
International Conference on Machine Learning (ICML)	*2020, 2021, 2022, 2023, 2024, 2025*
International Conference on Learning Representations (ICLR)	*2021, 2022, 2023, 2024, 2025*
Computer Vision and Pattern Recognition Conference (CVPR)	*2022, 2023, 2024, 2025*
International Conference on Computer Vision (ICCV)	*2023,2025*
European Conference on Computer Vision (ECCV)	*2022, 2024*
International Symposium on Information Theory (ISIT)	*2020, 2021*
Artificial Intelligence and Statistics (AISTATS)	*2021*
ICAIF Workshop on Synthetic Data for AI in Finance	*2022*
ACM CoNEXT NativeNI Workshop	*2022*
NeurIPS SyntheticData4ML Workshop	*2022*

Transactions on Machine Learning Research
Transactions on Pattern Analysis and Machine Intelligence
IET Image Processing
Transactions on Dependable and Secure Computing
Neural Networks
IEEE Access
IEEE/ACM Transactions on Networking
IEEE Transactions on Big Data
International Journal of Intelligent Systems

Publications (* denotes equal contribution, + denotes (co-)advising)

[1] **Zinan Lin**, Tadas Baltrusaitis, Wenyu Wang, and Sergey Yekhanin. "Differentially Private Synthetic Data via APIs 3: Using Simulators Instead of Foundation Models". In: ***ICLR Workshop on Data Problems, ICLR Workshop on Synthetic Data*** (2025). URL: https://arxiv.org/abs/2502.05505.

[2] Enshu Liu, Xuefei Ning, Yu Wang, and **Zinan Lin+**. "Distilled Decoding 1: One-step Sampling of Image Auto-regressive Models with Flow Matching". In: *International Conference on Learning Representations* ***(ICLR)*** (2025). URL: https://arxiv.org/abs/2412.17153.

[3] Enshu Liu*, Junyi Zhu*, **Zinan Lin+**, Xuefei Ning+, Matthew B Blaschko, Sergey Yekhanin, Shengen Yan, Guohao Dai, Huazhong Yang, and Yu Wang. "Linear Combination of Saved Checkpoints Makes Consistency and Diffusion Models Better". In: *International Conference on Learning Representations* ***(ICLR)*** (2025). URL: https://arxiv.org/abs/2404.02241.

[4] Chejian Xu, Jiawei Zhang, Zhaorun Chen, Chulin Xie, Mintong Kang, Zhuowen Yuan, Zidi Xiong, Chenhui Zhang, Lingzhi Yuan, Yi Zeng, Peiyang Xu, Chengquan Guo, Andy Zhou, Jeffrey Ziwei Tan, Zhun Wang, Alexander Xiong, Xuandong Zhao, Yu Gai, Francesco Pinto, Yujin Potter, Zhen Xiang, **Zinan Lin**, Dan Hendrycks, Dawn Song, and Bo Li. "MMDT: Decoding the Trustworthiness and Safety of Multimodal Foundation Models". In: *International Conference on Learning Representations* ***(ICLR)*** (2025). URL: https://arxiv.org/abs/2503.14827.

[5] Tianchen Zhao, Tongcheng Fang, Enshu Liu, Wan Rui, Widyadewi Soedarmadji, Shiyao Li, **Zinan Lin**, Guohao Dai, Shengen Yan, Huazhong Yang, Xuefei Ning, and Yu Wang. "ViDiT-Q: Efficient and Accurate Quantization of Diffusion Transformers for Image and Video Generation". In: *International Conference on Learning Representations* ***(ICLR)*** (2025). URL: https://arxiv.org/abs/2406.02540.

[6] Shuaiqi Wang, Shuran Zheng, **Zinan Lin**, Giulia Fanti, and Zhiwei Steven Wu. "Inferentially-Private Private Information". In: *The ACM Web Conference* ***(WWW)*** (2025). URL: https://arxiv.org/abs/2410.17095.

[7] Gong Chen, Kecen Li, **Zinan Lin**, and Wang Tianhao. "DPImageBench: A Unified Benchmark for Differentially Private Image Synthesis". In: *arXiv preprint arXiv:2503.14681* (2025). URL: https://arxiv.org/abs/2503.14681.

[8] Xuefei Ning*, Zifu Wang*, Shiyao Li*, **Zinan Lin***, Peiran Yao*, Tianyu Fu, Matthew B Blaschko, Guohao Dai, Huazhong Yang, and Yu Wang. "Can LLMs Learn by Teaching for Better Reasoning? A Preliminary Study". In: *Advances in Neural Information Processing Systems* ***(NeurIPS)*** (2024). URL: https://arxiv.org/abs/2406.14629.

[9] Sangyun Lee, **Zinan Lin**, and Giulia Fanti. "Improving the Training of Rectified Flows". In: *Advances in Neural Information Processing Systems* ***(NeurIPS)*** (2024). URL: https://arxiv.org/abs/2405.20320.

[10] Chengquan Guo*, Xun Liu*, Chulin Xie*, Andy Zhou, Yi Zeng, **Zinan Lin**, Dawn Song, and Bo Li. "RedCode: Risky Code Execution and Generation Benchmark for Code Agents". In: *Advances in Neural Information Processing Systems* ***(NeurIPS)*** (2024). URL: https://arxiv.org/abs/2411.07781.

[11] Kaiyi Huang, Yukun Huang, Xuefei Ning, **Zinan Lin**, Yu Wang, and Xihui Liu. "GenMAC: Compositional Text-to-Video Generation with Multi-Agent Collaboration". In: *arXiv preprint arXiv:2412.04440* (2024). URL: https://arxiv.org/abs/2412.04440.

[12] Enshu Liu*****, Junyi Zhu*****, **Zinan Lin+**, Xuefei Ning**+**, Matthew B Blaschko, Shengen Yan, Guohao Dai, Huazhong Yang, and Yu Wang. "Efficient Expert Pruning for Sparse Mixture-of-Experts Language Models: Enhancing Performance and Reducing Inference Costs". In: *arXiv preprint arXiv:2407.00945* (2024). URL: https://arxiv.org/abs/2407.00945.

[13] **Zinan Lin**, Sivakanth Gopi, Janardhan Kulkarni, Harsha Nori, and Sergey Yekhanin. "Differentially Private Synthetic Data via Foundation Model APIs 1: Images". In: *International Conference on Learning Representations **(ICLR)*** (2024). URL: https://arxiv.org/abs/2305.15560.

[14] Chulin Xie, **Zinan Lin**, Arturs Backurs, Sivakanth Gopi, Da Yu, Huseyin Inan, Harsha Nori, Haotian Jiang, Huishuai Zhang, Yin Tat Lee, Bo Li, and Sergey Yekhanin. "Differentially Private Synthetic Data via Foundation Model APIs 2: Text". In: *International Conference on Machine Learning **(ICML)*** (2024). URL: https://arxiv.org/abs/2403.01749.

[15] Xuefei Ning*****, **Zinan Lin***, Zixuan Zhou*****, Huazhong Yang, and Yu Wang. "Skeleton-of-Thought: Prompting LLMs for Efficient Parallel Generation". In: *International Conference on Learning Representations **(ICLR)*** (2024). URL: https://arxiv.org/abs/2307.15337.

[16] Shuaiqi Wang, **Zinan Lin**, and Giulia Fanti. "Statistic Maximal Leakage". In: *The IEEE International Symposium on Information Theory **(ISIT)*** (2024). URL: https://arxiv.org/abs/2411.18531.

[17] Xinyu Tang, Richard Shin, Huseyin A. Inan, Andre Manoel, Fatemehsadat Mireshghallah, **Zinan Lin**, Sivakanth Gopi, Janardhan Kulkarni, and Robert Sim. "Privacy-Preserving In-Context Learning with Differentially Private Few-Shot Generation". In: *International Conference on Learning Representations **(ICLR)*** (2024). URL: https://arxiv.org/abs/2309.11765.

[18] Arturs Backurs, **Zinan Lin**, Sepideh Mahabadi, Sandeep Silwal, and Jakub Tarnawski. "Efficiently Computing Similarities to Private Datasets". In: *International Conference on Learning Representations **(ICLR)*** (2024). URL: https://arxiv.org/abs/2403.08917.

[19] Junyi Zhu, **Zinan Lin**, Enshu Liu, Xuefei Ning, and Matthew B. Blaschko. "Rescaling Intermediate Features Makes Trained Consistency Models Perform Better". In: *International Conference on Learning Representations **(ICLR)** (TinyPapers)* (2024). URL: https://openreview.net/forum?id=1o3LnwflAI.

[20] Ronghao Ni, **Zinan Lin**, Shuaiqi Wang, and Giulia Fanti. "Mixture-of-Linear-Experts for Long-term Time Series Forecasting". In: *International Conference on Artificial Intelligence and Statistics **(AISTATS)*** (2024). URL: https://arxiv.org/abs/2312.06786.

[21] Lin Zhao*****, Tianchen Zhao*****, **Zinan Lin**, Xuefei Ning, Guohao Dai, Huazhong Yang, and Yu Wang. "FlashEval: Towards Fast and Accurate Evaluation of Text-to-image Diffusion Generative Models". In: *The IEEE/CVF Conference on Computer Vision and Pattern Recognition **(CVPR)*** (2024). URL: https://arxiv.org/abs/2403.16379.

[22] **Zinan Lin***, Shuaiqi Wang*****, Vyas Sekar, and Giulia Fanti. "Summary Statistic Privacy in Data Sharing". In: *IEEE Journal on Selected Areas in Information Theory **(JSAIT)*** (2024). URL: https://arxiv.org/abs/2303.02014.

[23] Da Yu, Sivakanth Gopi, Janardhan Kulkarni, **Zinan Lin**, Saurabh Naik, Tomasz Lukasz Religa, Jian Yin, and Huishuai Zhang. "Selective Pre-training for Private Fine-tuning". In: *Transactions on Machine Learning Research* ***(TMLR)*** (2024). URL: https://arxiv.org/abs/2305.13865.

[24] Tianchen Zhao, Xuefei Ning, Tongcheng Fang, Enshu Liu, Guyue Huang, **Zinan Lin**, Shengen Yan, Guohao Dai, and Yu Wang. "MixDQ: Memory-Efficient Few-Step Text-to-Image Diffusion Models with Metric-Decoupled Mixed Precision Quantization". In: *European Conference on Computer Vision* ***(ECCV)*** (2024). URL: https://arxiv.org/abs/2405.17873.

[25] Enshu Liu*****, Xuefei Ning*****, **Zinan Lin***, Huazhong Yang, and Yu Wang. "OMS-DPM: Optimizing the Model Schedule for Diffusion Probabilistic Models". In: *International Conference on Machine Learning* ***(ICML)***. 2023. URL: https://arxiv.org/abs/2306.08860.

[26] Boxin Wang, Weixin Chen, Hengzhi Pei, Chulin Xie, Mintong Kang, Chenhui Zhang, Chejian Xu, Zidi Xiong, Ritik Dutta, Rylan Schaeffer, Sang T Truong, Simran Arora, Mantas Mazeika, Dan Hendrycks, **Zinan Lin**, Yu Cheng, Sanmi Koyejo, Dawn Song, and Bo Li. "DecodingTrust: A Comprehensive Assessment of Trustworthiness in GPT Models". In: *Advances in Neural Information Processing Systems* ***(NeurIPS)***. 2023. URL: https://arxiv.org/abs/2306.11698.

[27] Da Yu, Arturs Backurs, Sivakanth Gopi, Huseyin Inan, Janardhan Kulkarni, **Zinan Lin**, Chulin Xie, Huishuai Zhang, and Wanrong Zhang. "Training Private and Efficient Language Models with Synthetic Data from LLMs". In: ***NeurIPS Workshop on Socially Responsible Language Modelling Research*** (2023). URL: https://openreview.net/forum?id=FKwtKzglFb.

[28] **Zinan Lin**. "Data Sharing with Generative Adversarial Networks: From Theory to Practice". PhD thesis. Carnegie Mellon University, 2022. URL: https://zinanlin.me/bio/PhD_thesis.pdf.

[29] **Zinan Lin***, Shuaiqi Wang*****, Vyas Sekar, and Giulia Fanti. "Distributional Privacy for Data Sharing". In: ***NeurIPS 2022 Workshop on Synthetic Data for Empowering ML Research***. URL: https://openreview.net/forum?id=6oVAzFsHLFK.

[30] Yucheng Yin, **Zinan Lin**, Minhao Jin, Giulia Fanti, and Vyas Sekar. "Practical GAN-based Synthetic IP Header Trace Generation using NetShare". In: *ACM Special Interest Group on Data Communication* ***(SIGCOMM)***. 2022. URL: https://dl.acm.org/doi/10.1145/3544216.3544251.

[31] **Zinan Lin**, Hao Liang, Giulia Fanti, and Vyas Sekar. "RareGAN: Generating Samples for Rare Classes". In: *AAAI Conference on Artificial Intelligence* ***(AAAI)***. 2022. URL: https://ojs.aaai.org/index.php/AAAI/article/view/20715.

[32] Yucheng Yin, **Zinan Lin**, Minhao Jin, Giulia Fanti, and Vyas Sekar. "PcapShare: Exploring the Feasibility of GANs for Synthetic Packet Header Trace Generation". In: *International Conference on COMmunication Systems and NETworkS* ***(COMSNETS)*** *(demo)*. 2022. URL: https://www.comsnets.org/demos_exhibits.html.

[33] **Zinan Lin**, Vyas Sekar, and Giulia Fanti. "Why Spectral Normalization Stabilizes GANs: Analysis and Improvements". In: *Advances in Neural Information Processing Systems* ***(NeurIPS)***. 2021. URL: http://arxiv.org/abs/2009.02773.

[34] **Zinan Lin**, Vyas Sekar, and Giulia Fanti. "On the Privacy Properties of GAN-generated Samples". In: *International Conference on Artificial Intelligence and Statistics* ***(AISTATS)***. PMLR. 2021, pp. 1522–1530. URL: https://arxiv.org/abs/2206.01349.

[35] Todd Huster, Jeremy E.J. Cohen, **Zinan Lin**, Kevin Chan, Cho-Yu Jason Chiang, and Vyas Sekar. "Pareto GAN: Extending the Representational Power of GANs to Heavy-Tailed Distributions". In: *International Conference on Machine Learning* ***(ICML)***. 2021. URL: http://proceedings.mlr.press/v139/huster21a.html.

[36] Mircea Trofin, Yundi Qian, Eugene Brevdo, **Zinan Lin**, Krzysztof Choromanski, and David Li. "MLGO: a Machine Learning Guided Compiler Optimizations Framework". In: *arXiv preprint arXiv:2101.04808* (2021). URL: https://arxiv.org/abs/2101.04808.

[37] **Zinan Lin**, Kiran Koshy Thekumparampil, Giulia Fanti, and Sewoong Oh. "InfoGAN-CR and ModelCentrality: Self-supervised Model Training and Selection for Disentangling GANs". In: *International Conference on Machine Learning* ***(ICML)***. 2020, pp. 7775–7786. URL: https://arxiv.org/abs/1906.06034.

[38] **Zinan Lin**, Alankar Jain, Chen Wang, Giulia Fanti, and Vyas Sekar. "Using GANs for Sharing Networked Timeseries Data: Challenges, Initial Promise, and Open Questions". In: *Proceedings of the Internet Measurement Conference* ***(IMC)***. 2020. URL: http://arxiv.org/abs/1909.13403.

[39] **Zinan Lin**, Ashish Khetan, Giulia Fanti, and Sewoong Oh. "PacGAN: The Power of Two Samples in Generative Adversarial Networks". In: *IEEE Journal on Selected Areas in Information Theory* ***(JSAIT)*** 1.1 (2020), pp. 324–335. URL: https://ieeexplore.ieee.org/document/9046238.

[40] **Zinan Lin**, Soo-Jin Moon, Carolina M. Zarate, Ritika Mulagalapalli, Sekar Kulandaivel, Giulia Fanti, and Vyas Sekar. "Towards Oblivious Network Analysis using Generative Adversarial Networks". In: *Proceedings of the 18th ACM Workshop on Hot Topics in Networks* ***(HotNets)***. ACM. 2019. URL: https://dl.acm.org/doi/10.1145/3365609.3365854.

[41] **Zinan Lin**, Ashish Khetan, Giulia Fanti, and Sewoong Oh. "PacGAN: The Power of Two Samples in Generative Adversarial Networks". In: *Advances in Neural Information Processing Systems* ***(NeurIPS)***. 2018, pp. 1498–1507. URL: https://arxiv.org/abs/1712.04086.

[42] Kiran K Thekumparampil, Ashish Khetan, **Zinan Lin**, and Sewoong Oh. "Robustness of Conditional GANs to Noisy Labels". In: *Advances in Neural Information Processing Systems* ***(NeurIPS)***. 2018, pp. 10271–10282. URL: https://arxiv.org/abs/1811.03205.

[43] **Zinan Lin**, Yongfeng Huang, and Jilong Wang. "RNN-SM: Fast Steganalysis of VoIP Streams Using Recurrent Neural Network". In: *IEEE Transactions on Information Forensics and Security* ***(TIFS)*** 13.7 (July 2018), pp. 1854–1868. ISSN: 1556-6013. DOI: 10.1109/TIFS.2018.2806741. URL: http://ieeexplore.ieee.org/document/8292900.